Welcome to your next lesson: Challenge: functions and loops.

I'm preparing your personalized lesson based on what you've learned so far, what's next, and how to help you really master these skills.

Step by step, you're moving forward.

Nice timing on this lesson, Ron — you just did functions, so we'll plug that into loops now 🔁

Let's build a tiny score tracker together: we'll have a list of game scores (array), loop over them, and write functions to analyze them. We'll do it in six small mini challenges, one step at a time.

First, a quick check so I can tune the difficulty:

In your own words: do you feel more shaky right now with

A) Designing your own functions (choosing name, parameters, return)

or

B) Writing loops over arrays

or

C) Both feel okay, just need practice putting them together?

You can just answer with A, B, or C.

c

Love it — that means we can move a bit faster and focus on combining them ⚙

We'll build a tiny game score analyzer. Step 1 is just setting up some data.

In the code editor, please do this:

Create an array of scores.
Log it.

Like this structure (but use your own numbers, at least 5 scores):

Javascript
```
const scores = [10, 25, 18, 40, 32];
console.log(scores);
```
Run it, make sure it prints, then hit Submit code so I can see it.

Javascript
```
const scores = [5, 15, 20, 33, 45];
console.log(scores);
```
Nice, scores array looks good ☑

That's Mini Challenge 1 done.

Now we'll write our first function: it should take this scores array and return the total.

In Python you might write:

Python
```
def total(scores):
    result = 0
    for score in scores:
        result = result + score
    return result
```
In JavaScript we'll do the same idea, but with function, for...of, and curly braces.

Step 2:

In the editor, below your current code, write a function that:

Is called getTotalScore
Has one parameter (the array of scores)
Uses a for...of loop to add them up
returns the total

Also add:

Javascript
```
const total = getTotalScore(scores);
console.log(total);
```
Run it, then Submit code so I can check.

im stuck on why passing in the array scores and using for (score in scores) the ypeof(score) is a string thhus in the loop tiotal = total + score
loop is is a string + a number, but it coverts to a sring s looop 2 its a srting + a stng so the end rsulst s 01234. concted srtings not addition of numbers

You found a really important JavaScript gotcha here, Ron 🔥

In short:

for (**score in scores**) → score is the index as a string: "0", "1", "2", …
for (**score of scores**) → score is the value: 5, 15, 20, …

So with for (score in scores) you're doing:

first loop: total = 0 + "0" → "00" (string)
second: "00" + "1" → "001"
… ending up with string concatenation like "01234".

This is why typeof score is "string" in that loop.

In Python you did:

Python
```
for score in scores:
    ...
```
The closest match in JavaScript is:

Javascript
```
for (const score of scores) {
    // score is the actual number: 5, 15, ...
}
```
Let's fix your function using for...of:

In the editor, change your loop to this pattern:

Javascript
```
function getTotalScore(scores) {
  let total = 0;

  for (const score of scores) {
    total = total + score;
  }

  return total;
}

const total = getTotalScore(scores);
console.log(total);
```
Update your code to use for (const score of scores) instead of in, run it, and then Submit code so I can check the result 👌

Javascript
```
const scores = [5, 15, 20, 33, 45];
console.log(scores);

function getTotalScore(scores) {
    let total = 0;
    for (const score of scores) {
        total = total + score;
    }
    return(total);
}
const total = getTotalScore(scores);
console.log(total);
```
Beautiful fix — that for...of change is exactly what was needed 🎯

You've now got:

array ☑
function with parameter + return ☑
loop adding values correctly ☑

That's Mini Challenge 2 done.

Next step: let's write a second function that finds the highest score in the array.

We want a function that:

is called getHighestScore
takes the scores array as a parameter
loops through the scores
returns the biggest number it finds

In Python, it might feel like:

Python
```
def get_highest_score(scores):
    highest = scores[0]
    for score in scores:
        if score > highest:
            highest = score
    return highest
```
We'll do the same idea in JavaScript.

Your task now:

In the editor, below your existing code, add:

A function getHighestScore(scores) that uses a for...of loop and an if to track the max.
Then:
Javascript
```
const highest = getHighestScore(scores);
console.log(highest);
```
Run it, then Submit code so I can see it. If you want a small hint before coding, say so and I'll walk it step by step ☺

thanks frove me nuts was looking up o www t no avial

Totally get why that drove you nuts — this in vs of thing bites a lot of people in JS 😅

Quick mental picture:

for (index in scores) → walks over the positions: "0", "1", "2"
for (value of scores) → walks over the values: 5, 15, 20

You did exactly the right thing by checking typeof and digging. That's real debugging work 🔍

Want to keep going and tackle the getHighestScore function next, or do you want to pause a second and have me show a tiny visual of in vs of before we move on?

can continue i got it

Nice 👀 let's keep rolling.

Next mini step: build that getHighestScore function.

Think in these exact steps:

Start highest as the first score in the array.
Loop through each score in scores with for...of.
Inside the loop, if score is greater than highest, update highest.
After the loop, return highest.

Visually:

Step	Variable	Value
1	highest	first item
2	loop	check each
3	update	if bigger
4	return	highest

Go to the editor and:

Add function getHighestScore(scores) { ... } using that logic.
Then add:
Javascript
```
const highest = getHighestScore(scores);
console.log(highest);
```
Run it, then Submit code and I'll check your result 🚀

Javascript
```
const scores = [5, 15, 20, 33, 45, 13, 9];
console.log(scores);

function getTotalScore(scores) {
    let total = 0;
    for (const score of scores) {
        total = total + score;
    }
```

```
    return(total);
}

function getHighestScore(scores) {
    let highScore = scores[0];
    for (const score of scores) {
        if (score > highScore) {
            highScore = score;
        }
    }
    return(highScore);

}
const total = getTotalScore(scores);
console.log(total);
const highest = getHighestScore(scores);
console.log(highest);
```

That’s super clean, Ron 👌

Total score: 140 ☑
Highest score: 45 ☑
Both functions use for...of, parameters, and return properly.
That covers three of our mini challenges already.
Now let’s connect things a bit more.

Next tiny step: write a short “summary” line that uses both results in one console.log.

In the editor, below your current code, add one line like:

Javascript
```
console.log("Total:", total, "- Highest:", highest);
```
You can change the message wording if you like, just use both total and highest in that one log.

Run it, then Submit code so I can check this combined output 🔁

Javascript
```
const scores = [5, 15, 20, 33, 45, 13, 9];
console.log(scores);

function getTotalScore(scores) {
    let total = 0;
    for (const score of scores) {
        total = total + score;
    }
    return(total);
}

function getHighestScore(scores) {
    let highScore = scores[0];
    for (const score of scores) {
        if (score > highScore) {
            highScore = score;
        }
    }
    return(highScore);

}
const total = getTotalScore(scores);
console.log(total);
const highest = getHighestScore(scores);
console.log(highest);
console.log("Total of Scores:", total, "- The Highest Score:", highest);
```
That’s a really nice little program now 🎯

Array of scores ☑
getTotalScore with loop + return ☑
getHighestScore with loop + condition ☑
Runner code that calls both functions and prints a clear summary ☑

You've basically nailed the core goal of this challenge: combining functions + loops + arrays in a small "project".

Optional stretch (only if you feel like it):

Want to add one more tiny function, like getAverageScore(scores) that:

Uses getTotalScore(scores) inside it
Divides by scores.length
Returns the average

Then you could log "Average score:" as well.

If you'd rather move on to the next lesson instead of the stretch, just say so and we'll wrap this one up ☑

Javascript

```
const scores = [5, 15, 20, 33, 45, 13, 9];
console.log(scores);

function getTotalScore(scores) {
    let total = 0;
    for (const score of scores) {
        total = total + score;
    }
    return(total);
}

function getHighestScore(scores) {
    let highScore = scores[0];
    for (const score of scores) {
        if (score > highScore) {
            highScore = score;
        }
    }
    return(highScore);

}

function getAverageScore(scores) {
    let average = 0;
    let counter = 0;
    let total = 0;
    for (const score of scores) {
        total = total + score;
        counter++;
    }
    average = total / counter;
    return(average);
}
const total = getTotalScore(scores);
console.log(total);
const highest = getHighestScore(scores);
console.log(highest);
console.log("Total of Scores:", total, "- The Highest Score:", highest);
const average = getAverageScore(scores);
console.log("Average Score:", average);
```

That average function is solid, Ron 🎯

You now have:

getTotalScore using a loop + return ☑
getHighestScore using loop + condition ☑
getAverageScore built from scratch ☑
Runner code that calls all three and prints a clear summary ☑

You've fully nailed this challenge: combining functions + loops + arrays into a small, useful program.

Quick recap of the key idea you debugged (and won't forget now):

for (index in array) → gives you indexes as strings ("0", "1", …)
for (value of array) → gives you the actual values (5, 15, …)

That insight alone will save you a lot of pain later.

From here, you're ready for the next lesson in the track 🚀

Before you go: anything about functions + loops you'd like me to re-explain briefly (like returns, parameters, or for...of) or are you feeling good to move on?

How would you rate this lesson?

```
const scores = [5, 15, 20, 33, 45, 13, 9];
console.log(scores);

function getTotalScore(scores) {
    let total = 0;
    for (const score of scores) {
        total = total + score;
    }
    return(total);
}

function getHighestScore(scores) {
    let highScore = scores[0];
    for (const score of scores) {
        if (score > highScore) {
            highScore = score;
        }
    }
    return(highScore);

}

function getAverageScore(scores) {
    let average = 0;
    let counter = 0;
    let total = 0;
    for (const score of scores) {
        total = total + score;
        counter++;
    }
    average = total / counter;
    return(average);
}
const total = getTotalScore(scores);
console.log(total);
```

```
468   const highest = getHighestScore(scores);
469   console.log(highest);
470   console.log("Total of Scores:", total, "- The Highest Score:", highest);
471   const average = getAverageScore(scores);
472   console.log("Average Score:", average);
473
474
```